

Probability Distribution functions :-

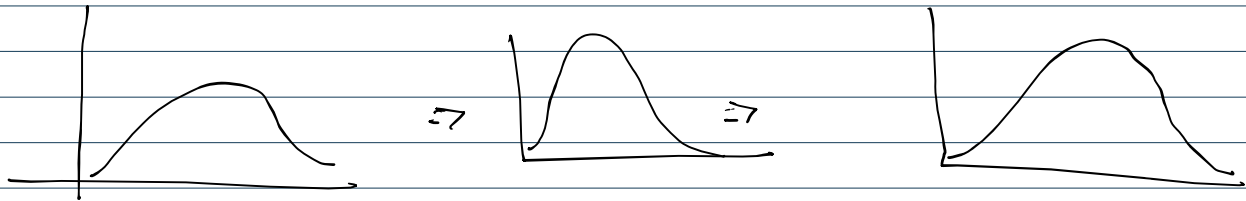
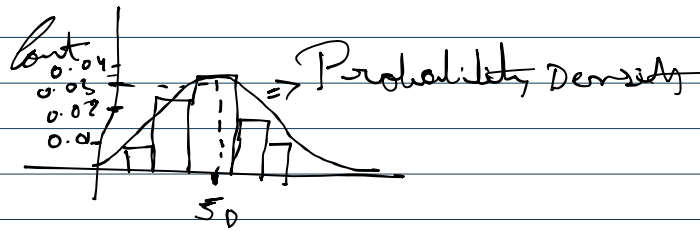
Probability distribution functions describe how the probabilities are distributed over the values of a Random Variable.

Ages = ξ - - - $\eta \Rightarrow$ Continuous random variables

$x \rightarrow F(x)$

$$\int_{-\infty}^{\infty} f(x) dx$$

Types



3 main topics

i) Probability Mass function (PMF) $\rightarrow$ discrete Random Variable

ii) Probability distribution function (PDF) $\rightarrow$ Continuous Random Variable

iii) Cumulative distribution function (CDF)

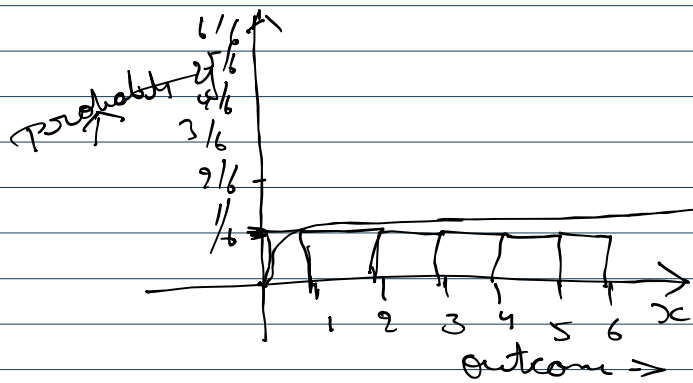
i) Probability Mass function

$\rightarrow$ Discrete Random Variable

Ex :- Rolling a dice $x = \begin{cases} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{cases}$

$$P_X(1) = \frac{1}{6} \quad P_X(2) = \frac{1}{6}$$

$$P_X(1) = P_X(2) = P_X(3) = P_X(4) = P_X(5) = P_X(6) = \frac{1}{6}$$



$n=7$

Cumulative density Function (cdf)

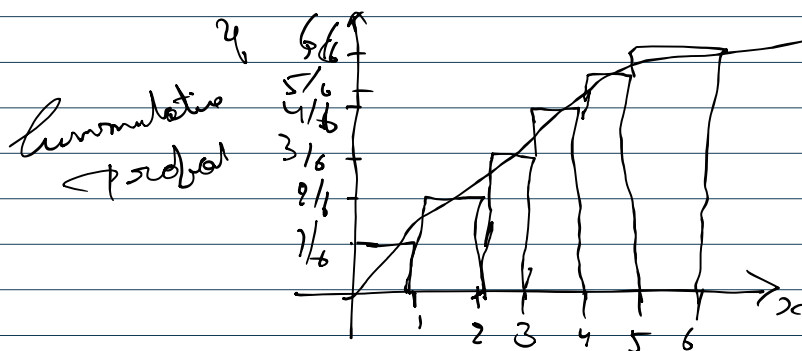
$$P(X=1) = 1/6$$

$x \rightarrow \text{Random } x \Rightarrow 1/6$

$x \leq 2$

$$\begin{aligned} P(X \leq 2) &= P(X=1) + P(X=2) \\ &\Rightarrow P(X=1) + P(X=2) \\ &= 1/6 + 1/6 \\ 2/6 &\leq 2/6 = 1/3 \end{aligned}$$

$$\begin{aligned} P(X \leq 3) &= P(X=1) + P(X=2) + P(X=3) \\ &= 1/6 + 1/6 + 1/6 \Rightarrow 3/6 \end{aligned}$$



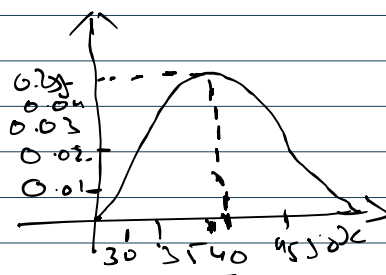
$P(X \leq 2)$

$P(X \leq 3)$

ii) Probability distribution function (PDF)

(1) Continuous Random

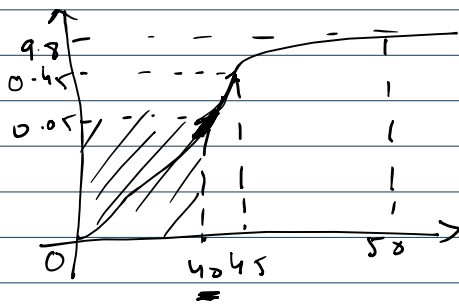
$x \Rightarrow \text{weights } \{ \dots \}$



Area under curve

probability density

$$P(X=40) \Rightarrow 0.05$$



$$P(X \leq 40) = 0.5$$

$$P(40 < X \leq 45)$$

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

Gradient $\Rightarrow$ Probability density.

Properties of PDF

i) Non-Negative $f(x) \geq 0$ for all x

(2) Total area under the PDF = 1

(3)

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$f(x)$



$$p + q = 1$$

$$q = 1 - p$$

Type of probability functions

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